

MENDIG, Germany

WMO: 105140

Lat: 50.366N

Lon: 7.315E

Elev: 182

StdP: 99.16

Time Zone: 1.00 (EUC)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.5	-7.0	-11.4	1.4	-8.3	-9.6	1.7	-5.3	12.7	7.8	11.2	7.5	1.5	240	0.530

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	31.6	19.8	29.6	19.2	27.6	18.5	21.0	29.0	20.1	27.7	19.1	25.9	3.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.5	23.4	17.4	12.7	22.6	16.5	12.0	22.0	61.3	29.2	58.2	27.6	55.0	25.9	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.0	7.0	-13.3	35.1	3.3	1.9	-15.7	36.5	-17.6	37.5	-19.4	38.6	-21.8	40.0		
(5)			-13.4	23.0	3.2	0.9	-15.8	23.6	-17.7	24.1	-19.5	24.6	-21.8	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	1.6	3.3	6.3	9.4	13.6	16.8	18.9	19.3	14.7	10.3	5.6	2.2	(6)
(7)		DBStd	7.17	4.60	4.15	3.61	3.58	3.50	3.49	3.26	3.33	2.82	3.45	3.40	4.32	(7)
(8)		HDD10.0	1052	261	188	123	54	7	0	0	0	1	39	136	242	(8)
(9)		HDD18.3	3109	518	421	372	268	151	71	33	28	114	251	382	499	(9)
(10)		CDD10.0	1128	1	1	9	36	119	205	275	289	142	47	3	1	(10)
(11)		CDD18.3	146	0	0	0	0	5	26	50	58	6	1	0	0	(11)
(12)		CDH23.3	1761	0	0	0	6	108	323	567	678	77	2	0	0	(12)
(13)		CDH26.7	607	0	0	0	0	21	95	206	268	17	0	0	0	(13)
(14)	Wind	WSAvg	2.8	3.3	3.4	3.1	2.9	2.6	2.6	2.6	2.7	2.8	2.6	3.0	(14)	
(15)	Precipitation	PrecAvg	684	51	48	48	44	61	69	70	64	60	54	56	60	(15)
(16)		PrecMax	852	125	110	105	87	131	141	164	137	148	127	103	145	(16)
(17)		PrecMin	540	9	9	9	1	12	22	28	10	22	16	5	19	(17)
(18)		PrecStd	83	24	24	24	20	27	28	31	33	33	25	22	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	14.2	19.8	24.2	29.2	32.1	33.7	35.2	29.3	22.8	14.9	12.7	(19)
(20)			MCWB	10.0	9.6	12.3	15.0	19.5	20.1	20.6	20.9	18.0	17.5	11.6	10.5	(20)
(21)		2%	DB	10.9	12.2	16.5	21.1	26.1	29.3	31.3	32.0	25.5	19.4	13.0	11.4	(21)
(22)			MCWB	8.8	8.8	10.9	13.3	17.4	19.6	19.7	20.0	17.5	15.0	10.5	9.5	(22)
(23)		5%	DB	9.3	10.8	14.2	18.7	24.0	27.2	29.0	29.8	22.9	17.4	11.7	10.1	(23)
(24)			MCWB	7.5	8.2	9.8	12.2	16.1	18.5	19.2	19.0	16.1	14.2	9.7	8.5	(24)
(25)		10%	DB	7.8	9.4	12.3	16.5	21.2	24.8	26.6	27.3	20.5	15.6	10.5	8.5	(25)
(26)			MCWB	6.2	7.2	9.1	11.1	14.7	17.4	18.2	18.4	15.0	12.8	8.9	7.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.5	13.2	16.3	20.2	22.0	22.1	22.1	19.7	18.0	12.3	10.9	(27)
(28)			MCDWB	12.2	13.0	17.8	22.2	27.0	29.2	31.0	30.7	26.3	22.2	13.8	12.1	(28)
(29)		2%	WB	8.9	9.3	11.6	14.1	18.1	20.4	20.7	20.9	18.0	15.7	11.0	9.7	(29)
(30)			MCDWB	10.6	11.8	15.4	19.4	25.7	27.4	28.7	29.5	24.1	18.7	12.6	11.2	(30)
(31)		5%	WB	7.6	8.4	10.5	12.7	16.5	19.1	19.8	20.0	16.8	14.4	9.9	8.5	(31)
(32)			MCDWB	9.2	10.5	13.4	17.5	22.6	26.2	27.3	27.8	21.7	16.9	11.4	9.9	(32)
(33)		10%	WB	6.2	7.4	9.3	11.7	15.2	17.9	18.8	19.0	15.6	13.1	8.9	7.2	(33)
(34)			MCDWB	7.6	9.3	12.0	15.8	20.3	23.6	25.1	25.5	19.5	15.4	10.4	8.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	7.1	8.9	10.8	11.3	12.1	11.7	12.2	10.3	8.8	6.2	5.4	(35)
(36)			MCDBR	6.9	8.8	12.7	15.1	16.4	16.6	16.6	17.0	15.1	11.3	7.8	5.9	(36)
(37)			MCWBR	5.5	6.0	7.9	8.3	8.7	8.0	7.1	6.9	7.7	7.4	5.8	4.8	(37)
(38)		5% WB	MCDBR	6.8	7.9	10.8	13.5	15.1	14.7	14.6	14.4	12.9	10.6	7.3	5.7	(38)
(39)			MCWBR	5.6	5.8	7.0	7.6	8.3	7.6	6.6	6.4	6.9	7.2	5.8	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.341	0.375	0.405	0.411	0.414	0.415	0.410	0.381	0.364	0.338	0.313	(40)	
(41)		taud	2.413	2.353	2.280	2.240	2.256	2.269	2.259	2.303	2.363	2.399	2.406	2.401	(41)	
(42)		Ebn at Noon	719	787	820	835	846	845	837	822	807	751	680	661	(42)	
(43)		Edh at Noon	63	85	108	126	129	129	128	117	99	81	63	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.77	1.46	2.60	4.14	4.86	5.37	5.08	4.36	3.24	1.89	0.87	0.59	(44)	
(45)		RadStd	0.12	0.28	0.40	0.49	0.47	0.51	0.59	0.43	0.41	0.19	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page